



STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

08 – Derivation of Wave-Induced Loads

This attachment provides a description of how the WA-OCEAN spreadsheet computes wave-induced loads for input into RISA-3D. Wave-induced loads result from (1) translation/rotation inertia and (2) dead load rotation. The RISA-3D program does not include this capability.

Inertia forces are determined by multiplying a mass by an acceleration ($F = M A$). The mass is determined by dividing a structure into individual components. The acceleration is determined from information provided by the Naval Architect. Note that inertia forces are opposite in direction to the acceleration.

Inertia Force = (component mass) $(-1 * \text{transportation acceleration})$ (F.1)

For calculation simplicity, a transformation is applied where the component mass is multiplied by the acceleration of gravity and the transportation acceleration is divided by the acceleration of gravity.

Inertia Force = (component weight) $(-1 * \text{acceleration coefficient})$ (F.2)

Inertia forces are further described in Sections 1 through 6.

Dead load rotation forces are described in Section 7.

1.0 COMPONENT DEFINITION APPROACH

There are two basic approaches to defining components.

One is to use a joint lumped inertia approach. This basically converts a defined set of vertical dead loads (nodal, point, and distributed loads) into a tributary weight at each node in the structure. The component weights are then the individual tributary weights. This is the same approach as the calculation of mass for a dynamic analysis.

The other is to use a member distributed inertia approach. This basically adapts a defined set of vertical dead loads (nodal, point, and distributed loads) while retaining their defined form. The component weights are then the nodal, point, and distributed loads. For a vertical downward $1g$ inertia force, this would be identical to applying the defined dead load directly.

On a global model scale, the methods should produce very similar results. On a local member scale, there is a difference in how inertia forces are applied between nodes. For simplicity and accuracy of local member checks, the WA-OCEAN spreadsheet uses the member distributed inertia approach.

2.0 DESIGN CASE APPROACH

There are two approaches to computing inertia force design cases. A design case is a specific set of surge, sway, heave, pitch, roll and yaw accelerations. The list of design cases is compiled for head sea, beam sea, and quartering sea conditions. A RISA-3D load combination will be required for each design case.

The simplest approach is to directly compute each design case. This will result in a one basic load case for each load combination. Each load combination will refer to the associated basic load case and apply a load factor of 1.0.

A second approach is to separately compute individual inertia forces (surge, sway, heave, pitch, roll, and yaw) based on nominal acceleration magnitudes. This will result in six basic load cases. Each load combination refers to all six basic load cases and applies load factors that will result in the associated design case.



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Either approach will result in the same inertia forces. The WA-OCEAN spreadsheet uses the second approach because it requires fewer computer resources.

3.0 ACCELERATION COEFFICIENTS

Marine loading is provided by the Naval Architect typically in terms of 3 translations and 3 rotations. Each of these six values could be described in terms of a peak value and period.

For translation motions, peak values and periods are typically pre-converted into an acceleration coefficient (refer to Section 4). Adapting equation F.2 and the LTV coordinates described in Appendix B results in:

$$\begin{aligned} \text{(translation along L axis)} \quad F_L &= -(CW)(TC_L) & (F.3) \\ \text{(translation along T axis)} \quad F_T &= -(CW)(TC_T) \\ \text{(translation along V axis)} \quad F_V &= -(CW)(TC_V) \end{aligned}$$

Where F = inertia force; CW = component weight; TC = translation acceleration coefficient

For rotational motions, peak values and periods are typically not pre-converted. Acceleration coefficients can be determined from Section 5. Adapting equation F.2 and the LTV coordinates described in Attachment 2 results in:

$$\begin{aligned} \text{(rotation about L axis)} \quad F_T &= -(CW)(-RC_L)(L_V) \quad F_V = -(CW)(+RC_L)(L_T) & (F.4) \\ \text{(rotation about T axis)} \quad F_L &= -(CW)(+RC_T)(L_V) \quad F_V = -(CW)(-RC_T)(L_L) \\ \text{(rotation about V axis)} \quad F_L &= -(CW)(-RC_V)(L_T) \quad F_T = -(CW)(+RC_V)(L_L) \end{aligned}$$

Summing equations F.3 and F.4 for F_L , F_T , and F_V results in:

$$\begin{aligned} F_L &= -(-CW)(RC_V) L_T + (-CW)(RC_T) L_V + (-CW)(TC_L) & (F.5) \\ F_T &= +(-CW)(RC_V) L_L - (-CW)(RC_L) L_V + (-CW)(TC_T) \\ F_V &= -(-CW)(RC_T) L_L + (-CW)(RC_L) L_T + (-CW)(TC_V) \end{aligned}$$

Re-organizing:

$$\begin{Bmatrix} F_L \\ F_T \\ F_V \end{Bmatrix} = -(CW) \left(\begin{bmatrix} (0) & (-RC_V) & (+RC_T) \\ (+RC_V) & (0) & (-RC_L) \\ (-RC_T) & (+RC_L) & (0) \end{bmatrix} \begin{Bmatrix} L_L \\ L_T \\ L_V \end{Bmatrix} + \begin{Bmatrix} TC_L \\ TC_T \\ TC_V \end{Bmatrix} \right) \quad (F.6)$$

After using Sections 4 and 5 to translate from LTV coordinates to XYZ coordinates, this becomes,

$$\begin{Bmatrix} F_X \\ F_Y \\ F_Z \end{Bmatrix} = -(CW) \left(\begin{bmatrix} (0) & (-RC_Z) & (+RC_Y) \\ (+RC_Z) & (0) & (-RC_X) \\ (-RC_Y) & (+RC_X) & (0) \end{bmatrix} \begin{Bmatrix} L_X \\ L_Y \\ L_Z \end{Bmatrix} + \begin{Bmatrix} TC_X \\ TC_Y \\ TC_Z \end{Bmatrix} \right) \quad (F.7)$$



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4.0 FACTORED NOMINAL LOADING

The WA-OCEAN spreadsheet computes nominal loads for RISA-3D input, with load factors applied later that amplify nominal loads to design level loads. Because of the resulting separation between the distribution of loading (in the nominal loads) and the overall magnitude of loading (in the factor), the approach allows a greater flexibility in the application of wave-induced loads.

The nominal loads are applied directly in the XYZ coordinate system used by RISA-3D.

A translation coefficient is applied in each of the three coordinate directions. Acceleration values are negative in order to produce a positive inertia force:

Nominal translation acceleration = -9.80665 m/s²

Nominal translation coefficient, TC = (-9.80665 m/s²) / (gravity) = -1.0

Using a prime mark to indicate nominal loads, the 3 translation results are described in equation form as,

X axis translation → $F_x = -(CW)(TC'x)$ (F.8)

Y axis translation → $F_y = -(CW)(TC'y)$

Z axis translation → $F_z = -(CW)(TC'z)$

A rotational coefficient is applied directly about each of the three coordinate directions.

Defined nominal rotation acceleration = -0.1 rad/s²

Nominal rotational coefficient = (-0.1 rad/s²) / (gravity) = -0.010197

Using a prime mark to indicate nominal loads, the 3 rotation results are described in equation form as,

X axis rotation → $F_y = (-CW)(-RC'x)(L_z)$ $F_z = (-CW)(+RC'x)(L_y)$ (F.9)

Y axis rotation → $F_x = (-CW)(+RC'y)(L_z)$ $F_z = (-CW)(-RC'y)(L_x)$

Z axis rotation → $F_x = (-CW)(-RC'z)(L_y)$ $F_y = (-CW)(+RC'z)(L_x)$

These results can be combined as,

$$\begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} = -(CW) \begin{bmatrix} (0) & (-RC'_z) & (+RC'_y) \\ (+RC'_z) & (0) & (-RC'_x) \\ (-RC'_y) & (+RC'_x) & (0) \end{bmatrix} \begin{Bmatrix} L_x \\ L_y \\ L_z \end{Bmatrix} + \begin{Bmatrix} TC'_x \\ TC'_y \\ TC'_z \end{Bmatrix} \quad (F.10)$$

Applying 6 factor variables as follows,

$$\begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} = -(CW) \begin{bmatrix} (0) & (F_6)(-RC'_z) & (F_5)(+RC'_y) \\ (F_6)(+RC'_z) & (0) & (F_4)(-RC'_x) \\ (F_5)(-RC'_y) & (F_4)(+RC'_x) & (0) \end{bmatrix} \begin{Bmatrix} L_x \\ L_y \\ L_z \end{Bmatrix} + \begin{Bmatrix} (F_1) TC'_x \\ (F_2) TC'_y \\ (F_3) TC'_z \end{Bmatrix}$$

This is equivalent to equation F.7 provided that,

$$F_1 = (TC_x) / (TC'x) \quad (F.11)$$

$$F_2 = (TC_y) / (TC'y)$$

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$$F3 = (TCz) / (TC'z)$$

$$F4 = (RCx) / (RC'x)$$

$$F5 = (RCy) / (RC'y)$$

$$F6 = (RCz) / (RC'z)$$

Computing a set of nominal values individually for each of the six modes allows the nominal results to be factored and combined as needed for any required load combination.

5.0 TRANSLATION ACCELERATION EQUATIONS

If point “Q” is subjected to a harmonic translation acceleration, the amplitude (at any point in time) is,

$$\Delta(t) = \Delta_M \sin(\omega t + \alpha) \quad (F.12)$$

Where,

$\Delta(t)$ = time-varying amplitude (m)

Δ_M = peak amplitude (m)

ω = circular frequency (rad/s) = $2\pi / T$

α = phase angle (rad)

t = time (s)

T = period (s)

The velocity is the first derivative of amplitude,

$$V(t) = \text{Velocity (m/s)} = \Delta'(t) = \Delta_M \omega \cos(\omega t + \alpha) \quad (F.13)$$

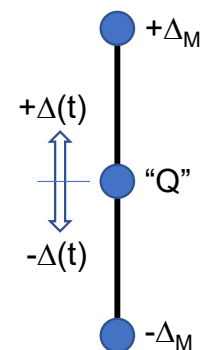
The acceleration is the first derivative of velocity,

$$A(t) = \text{Acceleration (m/s}^2\text{)} = V'(t) = -\Delta_M \omega^2 \sin(\omega t + \alpha) \quad (F.14)$$

Thus, peak acceleration values regardless of time are:

Peak acceleration, $A_M = -\Delta_M \omega^2$

Peak acceleration coefficient, $AC_M = -\Delta_M \omega^2 / g$



The resulting inertia force is the mass at point “Q” times the acceleration:

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$F(t) = -(\text{component mass} \times g)(\text{acceleration coefficient}) = (\text{component weight})(\text{acceleration coefficient})$

Peak force, $F_M = -(CW)(AC_M)$ (F.15)

Example using peak amplitude, $\Delta_M = +4$ m, period, $T = 10$ s, component weight, $CW = 15$ kN

Circular frequency, $\omega = 2\pi / T = (2\pi) / (10 \text{ s}) = 0.628 \text{ rad/s}$

Peak acceleration, $A_M = -\Delta_M \omega^2 = - (4 \text{ m})(0.628 \text{ rad/s})^2 = -1.58 \text{ m/s}^2$

Peak acceleration coefficient, $AC_M = -\Delta_M \omega^2 / g = - (1.58 \text{ m/s}^2) / (9.8 \text{ m/s}^2) = -0.161$

Peak force = $-(CW)(AC_M) = - (15 \text{ kN})(-0.161) = +2.42 \text{ kN}$

6.0 ROTATION ACCELERATION EQUATIONS

If point “CoR” is subjected to a harmonic rotational acceleration, the rotation (at any point in time) is.

$$\phi(t) = \phi_M \sin(\omega t + \alpha)$$

Where,

$\phi(t)$ = time-varying amplitude (radians)

ϕ_M = peak amplitude (radians)

ω = circular frequency (rad/s) = $2\pi / T$

α = phase angle (rad)

t = time (s)

T = period (s)

The resulting amplitude at rigidly connected point “Q”, at any point in time is,

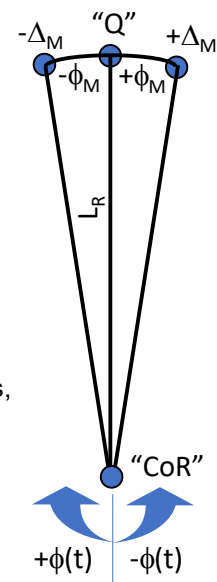
$$\Delta(t) = \Delta_M \sin(\omega t + \alpha) \quad (\text{F.16})$$

Where,

$\Delta(t)$ = time-varying tangential amplitude (m)

Δ_M = peak tangential amplitude (m)

The tangential amplitude Δ_M can be related to the amplitude ϕ_M using,





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L_R = radial distance between “CoR” and “Q” (m)

L_{CIRCLE} = full circle circumference (m) = $2 \pi L_R$

ϕ_{CIRCLE} = full circle angle (rad) = 2π

$$\Delta_M = (L_{CIRCLE}) (\phi_M / \phi_{CIRCLE}) = (2 \pi L_R) (\phi_M / 2 \pi) = L_R \phi_M \quad (F.17)$$

Following from the derivation of translational acceleration (Section 1),

circular acceleration (rad/s²), $A_C = - \phi_M \omega^2$

linear acceleration (m/s²), $A_L = - \Delta_M \omega^2 = - (L_R \phi_M)(2 \pi / T)^2 = - 4 \pi^2 L_R \phi_M / T^2$

separating out the distance term L_R leaves,

$$\text{linear acceleration coefficient, } AC = - 4 \pi^2 \phi_M / g T^2 \quad (F.18)$$

$$\text{peak force, } F_M = - (CW)(AC)(L_R) \quad (F.19)$$

Example using peak angle, $\phi_M = +20$ degrees, period, $T = 10$ s, distance, $L_R = 15$ m, and weight, $CW = 23$ kN

Peak angular amplitude, $\phi_M = +20$ degrees = $+0.349$ radians

circular frequency, $\omega = 2 \pi / T = (2 \pi) / (10 \text{ s}) = 0.628 \text{ rad/s}$

circular acceleration, $A_C = - \phi_M \omega^2 = - (0.349 \text{ radians})(0.628 \text{ rad/s})^2 = -0.138 \text{ rad/s}^2$

linear acceleration, $A_L = - 4 \pi^2 L_R \phi_M / T^2 = - 4 \pi^2 (15 \text{ m})(0.349 \text{ rad}) / (10 \text{ s})^2 = -2.067 \text{ m/s}^2$

linear acceleration coefficient, $AC = - 4 \pi^2 \phi_M / g T^2 = - 4 \pi^2 (0.349 \text{ rad}) / (9.8 \text{ m/s}^2)(10 \text{ s})^2 = -0.014 / \text{m}$

peak force = $-(CW)(AC)(L_R) = -(23 \text{ kN})(-0.014 / \text{m})(15 \text{ m}) = +4.83 \text{ kN}$

The preceding derived a tangential force at point “Q”. In other words, the component amplitude acceleration is perpendicular to a straight line from the center of rotation “CoR” to the component location “Q” (figure 1a below).

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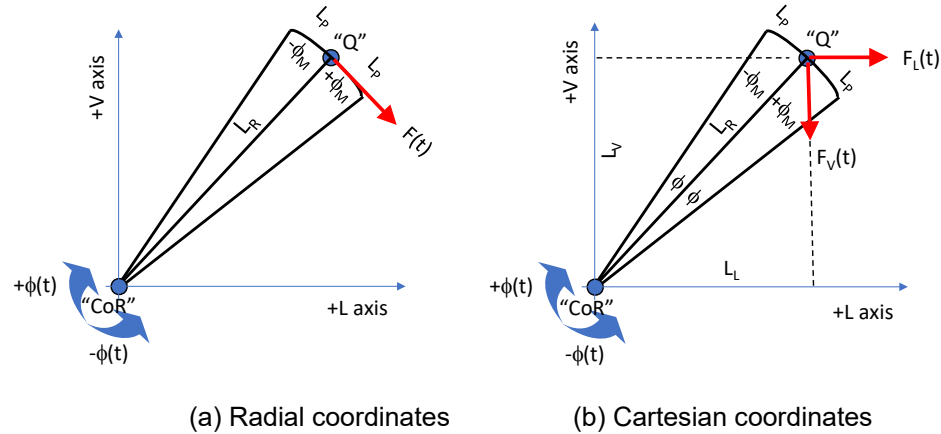


Figure 1: Acceleration Forces

An actual system using Cartesian coordinates will need a modification (Figure F1b above). The component force equation can then be modified accordingly.

$$F_{ML} = - (CW)(AC)(L_V) \quad (F.20)$$

$$F_{MV} = - (CW)(AC)(-L_L)$$

Continuing the previous example with $L_L = 12$ m and $L_V = 9$ m,

$$F_{ML} = - (CW)(AC)(L_V) = - (23 \text{ kN})(-0.014 / \text{m})(9 \text{ m}) = +2.90 \text{ kN}$$

$$F_{MV} = - (CW)(AC)(-L_L) = - (23 \text{ kN})(-0.014 / \text{m})(-12 \text{ m}) = -3.86 \text{ kN}$$

Confirming the result,

$$F_M = \text{sqrt} [(+2.90 \text{ kN})^2 + (-3.86 \text{ kN})^2] = +4.83 \text{ kN}$$

Note in Figure F1a that as the component “Q” change in position between $-\Delta_M$ and $+\Delta_M$, the tangential direction of the force also changes. This rotational effect is captured in the coordinate conversion described in Section B4.

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7.0 DEAD LOAD ROTATION

When computed relative to a marine vessel's deck, the magnitude of pitch and roll rotations leads to increased dead load lateral forces and a reduced dead load vertical force.

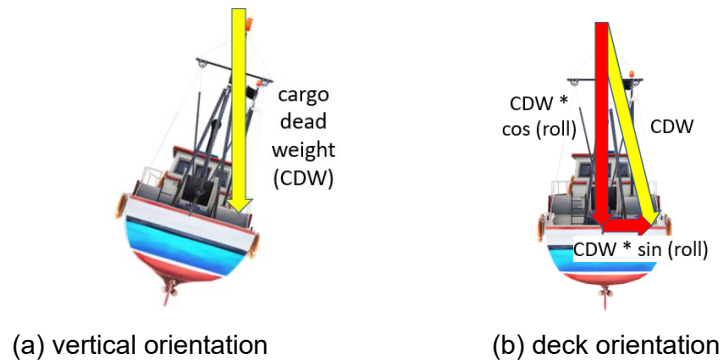


Figure 2: Rotation Effect on Dead Load

The WA-OCEAN spreadsheet uses dead load rotation equations based on the STAAD and SAP2000 approach. With the define LTV axis, the equations are,

$$(+\text{surge}) \text{ FL} = \text{CW} \times \sin (+\text{pitch}) \times -1 \text{ g} \quad (\text{F.21})$$

$$(+\text{sway}) \text{ FT} = \text{CW} \times \sin (-\text{roll}) \times -1 \text{ g}$$

$$(+\text{heave}) \text{ FV} = \text{CW} \times -1 \times \{ 1 - [\cos (\text{pitch}) \times \cos (\text{roll})] \}$$

Note that the heave value is the rotation effect on dead load. The dead load itself must still be applied. This distinction allows the application of different DNVGL/ISO code load factors for dead load and environmental load.